

Muhammad Adil Usmani

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SUMMARY

Software Engineer specializing in DL, Applicational AI, Retrieval-Augmented Generation (RAG) pipelines, and LLM workflow orchestration. Designs and integrates machine learning models served through FastAPI, managing unstructured text inputs and structured JSON data flows. Applies prompt engineering and context engineering to optimize LLM outputs across varied retrieval architectures.

TECHNICAL SKILLS

Languages: Python, SQL
Backend: FastAPI
AI & LLM: RAG, Prompt Engineering, Context Engineering, LLM Orchestration, LangChain, LangGraph, LangSmith, Azure OpenAI, OpenAI API, Gemini API, GPT-4o-mini, T5 Transformers
Databases: Neo4j, ChromaDB, FAISS, SQLite
Cloud & Tools: Azure, GCP, Render, Vercel, Claude, Claude Code, GitHub Copilot CLI, Digital Ocean, GitHub Actions, Git, Modal, Hugging Face

RESEARCH

Implementing Titan Architecture
Backend AI Research *Python, Titan Architecture, github.com/AadilUsmani/implementing_titan_architecture*

- Developed a backend infrastructure utilizing the Titan architecture to process and analyze large-scale financial datasets, including the Engro 2025 annual report.
- Integrated PDF data extraction and preprocessing pipelines to streamline the ingestion of complex corporate documents into the AI backend.

Lexical Graph RAG
Graph-Based Retrieval Research *github.com/AadilUsmani/Lexical_Graph_RAG, Python, Neo4j, LangGraph, Knowledge Graphs*

- Engineered a Graph-based Retrieval-Augmented Generation system using lexical graphs to enhance factual accuracy and contextual depth in AI-driven query responses.
- Implemented knowledge graph deduplication and extraction workflows to synthesize structured data from unstructured SEC filings and corporate texts.
- Evaluated performance using Signal-to-Noise and Cumulative Recall metrics to optimize the tradeoff between system latency and hallucination rates.

PROJECTS

AeroSphere – Air Quality Forecasting System
aerosphere.earth *Python, LSTM, Azure, Apache Airflow, GPT, NASA TEMPO*

- Engineered a 72-hour PM2.5 forecasting pipeline using LSTM models across 45 cities, ingesting satellite data from the NASA TEMPO dataset via Apache Airflow.
- Integrated GPT-based NLG to translate numerical model outputs into structured air quality reports; deployed full backend on Microsoft Azure.

Hybrid Graph RAG vs Vector RAG Evaluation System
github.com/AadilUsmani/V.G.RAG *Python, ChromaDB, Neo4j, Azure OpenAI*

- Built a hybrid RAG system querying Apple and Tesla SEC 10-K filings across ChromaDB (vector search) and Neo4j (graph traversal) with parallel dispatch pipelines.
- Designed a late-fusion aggregation layer and a 53-question automated benchmark to evaluate LLM accuracy, comprehensiveness, and directness.

Corrective RAG (CRAG) – Adaptive Routing API
github.com/AadilUsmani/Corrective_rag_CRAG *Python, LangGraph, FAISS, Tavily API, GPT-4o-mini*

- Architected a LangGraph stateful workflow with confidence-score-based routing: low-confidence queries (<0.3) trigger live web search; high-confidence (≥ 0.7) resolve from local FAISS store.

EDUCATION

University of Central Punjab <i>BS Computer Science</i>	Lahore, Pakistan
Unique Group of Institutions <i>Intermediate</i>	Lahore, Pakistan
Lahore Grammar School <i>Schooling</i>	Lahore, Pakistan

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification
- Introduction to Generative AI
- Machine Learning Specialization – Stanford University, Coursera (*In Progress*)
- LLMOps Specialization – Duke University, Coursera (*In Progress*)